# 📊 HR Workforce Analytics Dashboard

***Business Analyst Project Report***

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## 🎯 1. Objective

The objective of this project is to design and develop an **HR Employee Performance & Attendance Dashboard** using Excel to analyze workforce productivity, attendance patterns, and performance trends.

The dashboard enables HR teams to make **data-driven decisions** and improve organizational efficiency.

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## 🏢 2. Business Problem

Organizations face multiple workforce challenges such as:

- High absenteeism
- Low employee productivity
- Lack of performance visibility
- Inefficient department performance

Traditional reporting methods are static and do not provide actionable insights. This project solves these problems by building a **dynamic and interactive dashboard**.

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## 3. Dataset Description

The dataset includes the following key fields:

- Employee ID

- Employee Name
- Department
- Job Role
- Location
- Date & Month
- Attendance Status (Present, Absent, Leave, Work From Home)
- Working Hours
- Overtime Hours
- Tasks Completed
- Target Tasks
- Performance Score
- Salary
- Manager Rating

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# 📊 4. Data Processing

The data was cleaned and transformed by:

- Standardizing attendance values (P, A, L, WFH)
- Removing duplicates
- Formatting date and numerical values
- Handling missing values

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# 📈 5. Key Calculations

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## ✓ Productivity

Tasks Completed ÷ Target Tasks

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## ✓ Attendance %

Present Days ÷ Total Working Days

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### ✓ Achievement %

Tasks Completed ÷ Target Tasks

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### ✓ Performance Score

Based on ratings (1–10 scale)

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### ✓ Performance Categorization

Employees are categorized as:

- High Risk → Low performance + low attendance
- Performance Risk → Low performance
- Attendance Risk → Low attendance
- Stable → Good performance & attendance

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## 📊 6. Key Performance Indicators (KPIs)

- Total Employees
- Average Performance Score
- Attendance %
- Total Tasks Completed
- Avg Achievement %
- Bonus eligibility rate%
- Average management rate%
- Salary vs Achievement% correlation

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## 📊 7. Dashboard Features

The dashboard includes:

- Attendance status Distribution
- Avg overtime hours by job role
- Bonus eligibility rate by department
- Avg working hours by department
- Salary Distribution by Department
- Performance category distribution
- Avg Achievement % by department

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## KPI Cards

- Total Employee
- Departments
- Locations
- Present
- Absent
- WFH
- Avg.perf
- Avg.sal

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## Charts

- Pie Chart :  Attendance status Distridution & Performance Category Distribution.
- Horizontal Bar Chart : Average Working Hours by Department.
- Stacked Bar Chart: Average Overtime Hours by job role.
- Stacked column Chart : Bonus Eligibility Rate.
- Box & Whisker Plots :  Salary  Distribution by Department

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## Filters (Slicers)

- Department
- Location
- Month

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# 📉 8. Analysis & Insights

- Employees with higher attendance show better performance
- Certain departments have consistently higher productivity
- Some employees fall under high-risk category
- Overtime does not always lead to higher productivity
- High performers contribute significantly to overall output

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# 💼 9. Business Impact

This dashboard helps organizations to:

- Monitor workforce performance
- Reduce absenteeism
- Identify training needs
- Improve productivity
- Support HR decision-making

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# 🛠 10. Tools Used

- Microsoft Excel
- Pivot Tables
- Excel Formulas (SUMIFS, COUNTIFS, IF, XLOOKUP)
- Charts & Data Visualization

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# 🏁 11. Conclusion

This project demonstrates how Excel can be used as a powerful tool for **HR analytics**. The dashboard provides clear insights into employee performance and attendance, helping organizations make informed decisions.

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